

WEEK -1 PRACTICAL

USER COMMANDS

Id:

```
pavani@DESKTOP-380LEIN:~/ossp$ id
uid=1000(pavani) gid=1000(pavani) groups=1000(pavani),4(adm),24(cdrom),27(sudo),30(dip),46(plugdev),100(users)
```

Report:

After running the id command, I could see my user details. It showed my username as **pavani**, along with the user ID, group ID, and the groups I belong to like sudo and plugdev.

Env:

```

pavani@DESKTOP-380LEIN:~/ossp$ env
SHELL=/bin/bash
WSL2_GUI_APPS_ENABLED=1
WSL_DISTRO_NAME=Ubuntu
NAME=DESKTOP-380LEIN
PWD=/home/pavani/ossp
LOGNAME=pavani
HOME=/home/pavani
LANG=C.UTF-8
WSL_INTEROP=/run/WSL/443_interop
LS_COLORS=rs=0:di=01;34:ln=01;36:mh=00:pi=40;33:so=01;35:do=01;35:bd=40;33;01:cd=40;33;01:or=40;31;01:mi=00:su=37;41:sg=30;43:ca=00:tw=30;42:ow=34;42:st=37;44:ex=01;32:*.tar=01;31:*.tgz=01;31:*.arc=01;31:*.arj=01;31:*.taz=01;31:*.lha=01;31:*.lz4=01;31:*.lzh=01;31:*.lzma=01;31:*.tlz=01;31:*.txz=01;31:*.tzo=01;31:*.t7z=01;31:*.zip=01;31:*.z=01;31:*.dz=01;31:*.gz=01;31:*.lrz=01;31:*.lz=01;31:*.lzo=01;31:*.xz=01;31:*.zst=01;31:*.tzst=01;31:*.bz2=01;31:*.bz=01;31:*.tbz=01;31:*.tbz2=01;31:*.tz=01;31:*.deb=01;31:*.rpm=01;31:*.jar=01;31:*.war=01;31:*.ear=01;31:*.sar=01;31:*.rar=01;31:*.alz=01;31:*.ace=01;31:*.zoo=01;31:*.cpio=01;31:*.7z=01;31:*.rz=01;31:*.cab=01;31:*.wim=01;31:*.swm=01;31:*.dwm=01;31:*.esd=01;31:*.avif=01;35:*.jpg=01;35:*.jpeg=01;35:*.mjpg=01;35:*.mjpeg=01;35:*.gif=01;35:*.bmp=01;35:*.pbm=01;35:*.pgm=01;35:*.ppm=01;35:*.tga=01;35:*.xbm=01;35:*.xpm=01;35:*.tif=01;35:*.tiff=01;35:*.png=01;35:*.svg=01;35:*.svgz=01;35:*.mng=01;35:*.pcx=01;35:*.mov=01;35:*.mpg=01;35:*.mpeg=01;35:*.m2v=01;35:*.mkv=01;35:*.webm=01;35:*.webp=01;35:*.ogm=01;35:*.mp4=01;35:*.m4v=01;35:*.mp4v=01;35:*.vob=01;35:*.qt=01;35:*.nuv=01;35:*.wmv=01;35:*.asf=01;35:*.rm=01;35:*.rmvb=01;35:*.flc=01;35:*.avi=01;35:*.fli=01;35:*.flv=01;35:*.gl=01;35:*.dl=01;35:*.xcf=01;35:*.xwd=01;35:*.yuv=01;35:*.cgm=01;35:*.emf=01;35:*.ogv=01;35:*.ogx=01;35:*.aac=00;36:*.au=00;36:*.flac=00;36:*.m4a=00;36:*.mid=00;36:*.midi=00;36:*.mka=00;36:*.mp3=00;36:*.mpc=00;36:*.ogg=00;36:*.ra=00;36:*.wav=00;36:*.oga=00;36:*.opus=00;36:*.spx=00;36:*.xspf=00;36:*.~=00;90:*.bak=00;90:*.old=00;90:*.orig=00;90:*.part=00;90:*.rej=00;90:*.swp=00;90:*.tmp=00;90:*.dpkg-dist=00;90:*.dpkg-old=00;90:*.ucf-dist=00;90:*.ucf-new=00;90:*.ucf-old=00;90:*.rpmnew=00;90:*.rpmorig=00;90:*.rpmsave=00;90:
WAYLAND_DISPLAY=wayland-0
LESSCLOSE=/usr/bin/lesspipe %s %s
TERM=xterm-256color
LESSOPEN=| /usr/bin/lesspipe %s
USER=pavani
DISPLAY=:0
SHLVL=1
XDG_RUNTIME_DIR=/run/user/1000
WLENV=
XDG_DATA_DIRS=/usr/local/share:/usr/share:/var/lib/snapd/desktop
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/Program Files/WindowsApps/CanonicalGroupLimited.Ubuntu_2404.1.68.0_x64__79rhkp1fndgsc:/mnt/c/Program Files/Common Files/Oracle/Java/javapath:/mnt/c/Windows/system32:/mnt/c/Windows:/mnt/c/Windows/System32/Wbem:/mnt/c/Windows/System32/WindowsPowerShell/v1.0:/mnt/c/Windows/System32/OpenSSH:/mnt/c/Program Files/nodejs:/mnt/c/Users/USER/AppData/Local/Programs/Python/Python314/Scripts:/mnt/c/Users/USER/AppData/Local/Programs/Python/Python314:/mnt/c/Users/USER/AppData/Local/Programs/Python/Launcher:/mnt/c/Users/USER/AppData/Local/Microsoft/WindowsApps:/mnt/c/Users/USER/AppData/Local/Programs/Microsoft VS Code/bin:/mnt/c/Users/USER/AppData/Roaming/npm:/snap/bin
DBUS_SESSION_BUS_ADDRESS=unix:path=/run/user/1000/bus
HOSTTYPE=x86_64
PULSE_SERVER=unix:/mnt/wslg/PulseServer
OLDPWD=/home/pavani
_=/usr/bin/env

```

Report:

When I executed the env command, many environment variables were displayed. I observed variables like HOME=/home/pavani, USER=pavani, PWD=/home/pavani/OSSP, and PATH, which contain information about the system environment.

Export:

```

pavani@DESKTOP-380LEIN:~/ossp$ export
declare -x DBUS_SESSION_BUS_ADDRESS="unix:path=/run/user/1000/bus"
declare -x DISPLAY=":0"
declare -x HOME="/home/pavani"
declare -x HOSTTYPE="x86_64"
declare -x LANG="C.UTF-8"
declare -x LESSCLOSE="/usr/bin/lesspipe %s %s"
declare -x LESSOPEN="| /usr/bin/lesspipe %s"
declare -x LOGNAME="pavani"
declare -x LS_COLORS="rs=0:di=01;34:ln=01;36:mh=00:pi=40;33:so=01;35:do=01;35:bd=40;33;01:cd=40;33;01:or=40;31;01:mi=00:su=37;41:sg=30;43:ca=00:tw=30;42:ow=34;42:st=37;44:ex=01;32:*.tar=01;31:*.tgz=01;31:*.arc=01;31:*.arj=01;31:*.taz=01;31:*.lha=01;31:*.lz4=01;31:*.lzh=01;31:*.lzma=01;31:*.tlz=01;31:*.txz=01;31:*.tzo=01;31:*.t7z=01;31:*.zip=01;31:*.z=01;31:*.dz=01;31:*.gz=01;31:*.lrz=01;31:*.lz=01;31:*.lzo=01;31:*.xz=01;31:*.zst=01;31:*.tzst=01;31:*.bz2=01;31:*.bz=01;31:*.tbz=01;31:*.tbz2=01;31:*.taz=01;31:*.deb=01;31:*.rpm=01;31:*.jar=01;31:*.war=01;31:*.ear=01;31:*.sar=01;31:*.rar=01;31:*.alz=01;31:*.ace=01;31:*.zoo=01;31:*.cpio=01;31:*.7z=01;31:*.rz=01;31:*.cab=01;31:*.wim=01;31:*.swm=01;31:*.dwm=01;31:*.esd=01;31:*.avif=01;35:*.jpg=01;35:*.jpeg=01;35:*.mjpg=01;35:*.mjpeg=01;35:*.gif=01;35:*.bmp=01;35:*.pbm=01;35:*.pgm=01;35:*.ppm=01;35:*.tga=01;35:*.xbm=01;35:*.xpm=01;35:*.tif=01;35:*.tiff=01;35:*.png=01;35:*.svg=01;35:*.svgz=01;35:*.mng=01;35:*.pcx=01;35:*.mov=01;35:*.mpg=01;35:*.mpeg=01;35:*.m2v=01;35:*.mkv=01;35:*.webm=01;35:*.webp=01;35:*.ogm=01;35:*.mp4=01;35:*.m4v=01;35:*.mp4v=01;35:*.vob=01;35:*.qt=01;35:*.nuv=01;35:*.wmv=01;35:*.asf=01;35:*.rm=01;35:*.rmvb=01;35:*.flc=01;35:*.avi=01;35:*.fli=01;35:*.flv=01;35:*.gl=01;35:*.dl=01;35:*.xcf=01;35:*.xwd=01;35:*.yuv=01;35:*.cgm=01;35:*.emf=01;35:*.ogv=01;35:*.ogx=01;35:*.aac=00;36:*.au=00;36:*.flac=00;36:*.m4a=00;36:*.mid=00;36:*.midi=00;36:*.mka=00;36:*.mp3=00;36:*.mpc=00;36:*.ogg=00;36:*.ra=00;36:*.wav=00;36:*.oga=00;36:*.opus=00;36:*.spx=00;36:*.xspf=00;36:*.~=00;90:*.#=00;90:*.bak=00;90:*.old=00;90:*.orig=00;90:*.part=00;90:*.rej=00;90:*.swp=00;90:*.tmp=00;90:*.dpkg-dist=00;90:*.dpkg-old=00;90:*.ucf-dist=00;90:*.ucf-new=00;90:*.ucf-old=00;90:*.rpmnew=00;90:*.rpmorig=00;90:*.rpmsave=00;90:"
declare -x NAME="DESKTOP-380LEIN"
declare -x OLDPWD="/home/pavani"
declare -x PATH="/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/Program Files/WindowsApps/CanonicalGroupLimited.Ubuntu_2404.1.68.0_x64_79rhkp1fndgsc:/mnt/c/Program Files/Common Files/Oracle/Java/javapath:/mnt/c/Windows/system32:/mnt/c/Windows/mnt/c/Windows/System32/Wbem:/mnt/c/Windows/System32/WindowsPowerShell/v1.0:/mnt/c/Windows/System32/OpensH:/mnt/c/Program Files/nodejs:/mnt/c/Users/USER/AppData/Local/Programs/Python/Python314/Scripts:/mnt/c/Users/USER/AppData/Local/Programs/Python/Python314:/mnt/c/Users/USER/AppData/Local/Programs/Python/Launcher:/mnt/c/Users/USER/AppData/Local/Microsoft/WindowsApps:/mnt/c/Users/USER/AppData/Local/Programs/Microsoft VS Code/bin:/mnt/c/Users/USER/AppData/Roaming/npm:/snap/bin"
declare -x PULSE_SERVER="unix:/mnt/wslg/PulseServer"
declare -x PWD="/home/pavani/ossp"
declare -x SHELL="/bin/bash"
declare -x SHLVL="1"
declare -x TERM="xterm-256color"
declare -x USER="pavani"
declare -x WAYLAND_DISPLAY="wayland-0"
declare -x WSL2_GUI_APPS_ENABLED="1"
declare -x WSLENV=""
declare -x WSL_DISTRO_NAME="Ubuntu"
declare -x WSL_INTEROP="/run/WSL/443_interop"
declare -x XDG_DATA_DIRS="/usr/local/share:/usr/share:/var/lib/snapd/desktop"
declare -x XDG_RUNTIME_DIR="/run/user/1000"

```

Report: The export command displayed the exported environment variables. I observed that variables such as HOME, USER, PWD, PATH, and other system variables were available and exported for the current session

Alias:

```
pavani@DESKTOP-380LEIN:~/ossp$ alias
alias alert='notify-send --urgency=low -i "${[ $? = 0 ]}&& echo terminal || echo error)" "${history|tail
-n1|sed -e '\''s/^\s*[0-9]\+\s*//;s/[;&]\s*alert$//'\`''"'
alias egrep='egrep --color=auto'
alias fgrep='fgrep --color=auto'
alias grep='grep --color=auto'
alias l='ls -CF'
alias la='ls -A'
alias ll='ls -aF'
alias ls='ls --color=auto'
```

Report: After executing the alias command, the terminal displayed a list of shortcut commands. I observed aliases like grep, ls, and ll, which are used as shortcuts for commonly used commands

Task 1: Develop a C program that

demonstrates how a Linux operating system executes a command entered by a user that 1. Accept a Linux command as input. 2. Create a child process using fork(). 3. Execute the command in the child process using an appropriate exec() system call. 4. Allow the parent process to wait for the child using wait (). 5. Display the Process ID

CODE:

pavani@DESKTOP-38OLEIN: ~

GNU nano 8.7.1

command_executor.c

```
#include <stdio.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <string.h>

int main() {
    char command[100];
    pid_t pid;

    printf("Enter a Linux command: ");
    scanf("%s", command);

    pid = fork();

    if (pid < 0) {
        printf("Fork failed!\n");
        return 1;
    }

    else if (pid == 0) {
        printf("\nChild Process\n");
        printf("Child PID : %d\n", getpid());
        printf("Parent PID: %d\n", getppid());

        printf("\nExecuting command: %s\n\n", command);

        execlp(command, command, NULL);

        printf("Command execution failed!\n");
    }

    else {
        printf("\nParent Process\n");
        printf("Parent PID : %d\n", getpid());
        printf("Child PID : %d\n", pid);

        wait(NULL);

        printf("\nChild process completed.\n");
    }

    return 0;
}
```

OUTPUT:

```
pavani@DESKTOP-380LEIN: ~  
pavani@DESKTOP-380LEIN:~$ nano command_executor.c  
pavani@DESKTOP-380LEIN:~$ gcc command_executor.c -o command_executor  
pavani@DESKTOP-380LEIN:~$ ./command_executor  
Enter a Linux command: ls  
  
Parent Process  
Parent PID : 546  
Child PID : 547  
  
Child Process  
Child PID : 547  
Parent PID: 546  
  
Executing command: ls  
  
OSSP  command_executor  file2.txt  hello.sh      os-lab  shortcut  txt  
a.txt  command_executor.c  hello      helloCopy.java  ossp    snap  
b.txt  file1.txt            hello.c    main.java      script.sh  test.txt  
  
Child process completed.
```

Child process completed.

REPORT: I executed the C program successfully. The program accepted a Linux command from the user, created a child process using for The uname -a command displayed complete information about my Linux system, including the kernel version, system architecture, hostname, and operating system details.k(), and executed the command using the exec() system call. The parent process waited until the child process finished execution. Finally, the program displayed the process IDs of both the parent and child processes, and the message "Child process completed." was displayed.

Task-2: Using Linux terminal commands (uname, lscpu, ps, top), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices.

Uname-a:

```
pavani@DESKTOP-380LEIN:~$ uname -a  
Linux DESKTOP-380LEIN 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 18 21:54:43 UTC 2  
026 x86_64 GNU/Linux
```

REPORT: The uname -a command displayed complete information about my Linux system, including the kernel version, system architecture, hostname, and operating system details.

Lscpu:

```
pavani@DESKTOP-380LEIN: ~
026 x86_64 GNU/Linux
pavani@DESKTOP-380LEIN:~$ lscpu
Architecture:                x86_64
CPU op-mode(s):              32-bit, 64-bit
Address sizes:               39 bits physical, 48 bits virtual
Byte Order:                  Little Endian
CPU(s):                      8
On-line CPU(s) list:        0-7
Vendor ID:                   GenuineIntel
Model name:                  11th Gen Intel(R) Core(TM) i5-1135G7 @ 2.40GHz
CPU family:                  6
Model:                      140
Thread(s) per core:         2
Core(s) per socket:         4
Socket(s):                   1
Stepping:                    1
BogoMIPS:                    4838.39
Flags:                       fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clf
                             lush mmx fxsr sse sse2 ss ht syscall nx pdpe1gb rdtscp lm constant_tsc arch
                             _perfmmon rep_good nopl xtopology tsc_reliable nonstop_tsc cpuid tsc_known_f
                             req pni pclmulqdq vmx ssse3 fma cx16 pdcm pcid sse4_1 sse4_2 x2apic movbe p
                             opcnt tsc_deadline_timer aes xsave avx f16c rdrand hypervisor lahf_lm abm 3
                             dnowprefetch ssbd ibrs ibpb stibp ibrs_enhanced tpr_shadow ept vpid ept_ad
                             fsgsbase tsc_adjust bmi1 avx2 smep bmi2 erms invpcid avx512f avx512dq rdsee
                             d adx smap avx512ifma clflushopt clwb avx512cd sha_ni avx512bw avx512vl xsa
                             veopt xsavec xgetbv1 xsaves vnmi avx512vbmi umip avx512_vbmi2 gfni vaes vpc
                             lmulqdq avx512_vnni avx512_bitalg avx512_vpopcntdq rdpid movdiri movdir64b
                             fsrm avx512_vp2intersect md_clear ibt flush_l1d arch_capabilities

Virtualization features:
  Virtualization:            VT-x
  Hypervisor vendor:         Microsoft
  Virtualization type:       full
Caches (sum of all):
  L1d:                       192 KiB (4 instances)
  L1i:                       128 KiB (4 instances)
  L2:                         5 MiB (4 instances)
  L3:                         8 MiB (1 instance)
NUMA:
  NUMA node(s):              1
  NUMA node0 CPU(s):        0-7
Vulnerabilities:
  Gather data sampling:      Unknown: Dependent on hypervisor status
  Ghostwrite:                Not affected
  Indirect target selection: Mitigation; Aligned branch/return thunks
  Itlb multihit:            Not affected
  L1tf:                      Not affected
  Mds:                       Not affected
  Meltdown:                  Not affected
  Mmio stale data:          Not affected
  Old microcode:             Not affected
  Reg file data sampling:    Not affected
  Spectre_v1:                Mitigation; Enhanced IBPB
```

REPORT: The lscpu command displayed the CPU information of my system, such as the processor architecture, number of CPU cores, CPU model, and other hardware details. This helped me understand the processor resources available.

Ps:

```
pavani@DESKTOP-380LEIN:~$ ps
  PID TTY          TIME CMD
  363 pts/0        00:00:00 bash
  636 pts/0        00:00:00 ps
pavani@DESKTOP-380LEIN:~$
```

REPORT: The ps command displayed the processes currently running in my terminal. I observed the process IDs and command names of the active processes.

Top:

```
pavani@DESKTOP-380LEIN:~$ top
top - 09:05:15 up 7 min, 1 user, load average: 0.05, 0.05, 0.01
Tasks: 28 total, 1 running, 27 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni,100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 3803.6 total, 2402.1 free, 555.3 used, 982.8 buff/cache
MiB Swap: 1024.0 total, 1024.0 free, 0.0 used. 3248.3 avail Mem

  PID USER      PR  NI   VIRT   RES   SHR S  %CPU  %MEM    TIME+  COMMAND
    1 root        20   0   24384  15472 11616 S   0.0   0.4   0:00.76 systemd
    2 root        20   0    3180   2208  2072 S   0.0   0.1   0:00.01 init-systemd(Ub
    6 root        20   0    3180   2040   0.0 S   0.0   0.1   0:00.00 init
   46 root       19  -1   50396  16892 15592 S   0.0   0.4   0:00.21 systemd-journal
   77 systemd+   20   0   22540  14656 12032 S   0.0   0.4   0:00.07 systemd-resolve
   82 root       20   0   35560  12588  9316 S   0.0   0.3   0:00.19 systemd-udev
  144 root       20   0  159348   1836  1508 S   0.0   0.0   0:00.00 snapfuse
  150 root       20   0  454144  11612  1556 S   0.0   0.3   0:02.29 snapfuse
  177 root       20   0    2888   2016  1884 S   0.0   0.1   0:00.05 chronyd-starter
  178 root       20   0    4472   3168  2924 S   0.0   0.1   0:00.01 cron
  179 message+   20   0    8816   5508  4824 S   0.0   0.1   0:00.05 dbus-daemon
  183 root       20   0   44092  29820 14676 S   0.0   0.8   0:00.49 networkd-dispat
  186 root       20   0  2220748  38668 25020 S   0.0   1.0   0:00.85 snapd
  187 root       20   0   18444   9544  8292 S   0.0   0.2   0:00.07 systemd-logind
  190 root       20   0 1857652  15260 12616 S   0.0   0.4   0:00.15 wsl-pro-service
  207 syslog    20   0  220548   6020  4668 S   0.0   0.2   0:00.06 rsyslogd
  210 root       20   0    5492   2828  2644 S   0.0   0.1   0:00.01agetty
  222 root       20   0  123456  32588 17976 S   0.0   0.8   0:00.41unattended-upgr
  228 _chrony    20   0   24252  11904  7304 S   0.0   0.3   0:00.10 chronyd
  249 _chrony    20   0   12092   2512  1868 S   0.0   0.1   0:00.01 chronyd
  361 root       20   0    3188   1112   980 S   0.0   0.0   0:00.00 SessionLeader
  362 root       20   0    3204   1248  1108 S   0.0   0.0   0:00.01 Relay(363)
  363 pavani     20   0    6268   5600  3928 S   0.0   0.1   0:00.06 bash
  364 root       20   0    8648   5576  4732 S   0.0   0.1   0:00.01 login
  432 pavani    20   0   22340  13656 11032 S   0.0   0.4   0:00.10 systemd
  434 pavani    20   0   22912   4080  2076 S   0.0   0.1   0:00.00 (sd-pam)
  463 pavani    20   0    6136   5548  3992 S   0.0   0.1   0:00.02 bash
  646 pavani    20   0    8196   6084  3960 R   0.0   0.2   0:00.02 top
```

REPORT: The top command displayed the live status of the system. I observed the CPU usage, memory usage, running processes, and system performance updating continuously.